CONDENSEDMATH

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Contents

1. Condensed sets

Definition 1.0.1. A *profinite set* is a set which is the projective limit of a (small) directed system of finite sets.

A profinite set $X = \varprojlim_i X_i$ has a natural topology, referred to as the *profinite topology* or *inverse limit topology*. It can be formulated in the following equivalent ways:

(♠ TODO: define inverse limit topology)

1. As the inverse limit topology with respect to the family of projection maps $\pi_i : X \to X_i$ onto the finite sets X_i, each equipped with the discrete topology, in the system.
2. As the subspace topology on $\varprojlim_i X_i$ inherited from the product topology on $\prod_i X_i$.
3. As the topology generated by the basis of "finite index" sets (specifically, the clopen sets) of the form $\pi_i^{-1}(V)$ for some index i and some subset $V \subseteq X_i$.

These topologies do not depend on the inverse system representing X.

Proposition 1.0.2. Let X be a topological space. The following conditions are equivalent:

- X is a profinite set (Definition 1.0.1).
- X is compact, Hausdorff, and totally disconnected.
- X is a Stone space.

Definition 1.0.3. Let S and T be profinite sets (Definition 1.0.1). A *morphism $S \to T$ of profinite sets S and T* can be characterized in the following equivalent ways:

1. A continuous map between the compact Hausdorff totally disconnected topological spaces.
2. ... (♠ TODO: how to characterize directly on the inverse limit presentation)

Definition 1.0.4. The *category of profinite sets*, denoted by notations such as Prof, is the category whose objects are profinite sets (Definition 1.0.1) and whose morphisms are morphisms of profinite sets (Definition 1.0.3).

This category is equivalent to the category of compact Hausdorff totally disconnected topological spaces whose morphsims are continuous maps.

This category is also equivalent to the pro-category of the category of finite sets (♠ TODO: category of finite sets).

Prof admits all small limits.

Definition 1.0.5 ([CS22, Definition 2.2.], [Sch19, Definition 1.1.])**.** Define the following Grothendieck topology on the category Prof (Definition 1.0.1) of profinite sets (Definition 1.0.1): a finite collection $(f_i : S_i \to S)_{i\in I}$ of maps in Prof is a covering if the maps are jointly surjective, i.e. $\coprod_{i\in I} S_i \to S$ is surjective.

One may also call Prof equipped with this Grothendieck topology the *pro-étale site of the point*, and denote this site by $*_{\text{proét}}$.

It should be noted that Prof is not essentially small. (♠ TODO: talk about how the grothendieck topology here is finitary)

Definition 1.0.6 ([CS22, Definition 2.3.])**.** Let κ be a cardinal. A profinite set (Definition 1.0.1) S is called *κ-small* if the set of clopen subsets of S has cardinality $< \kappa$. The full subcategory of κ-small profinite sets is denoted $\text{Prof}_\kappa \subset \text{Prof}$ (Definition 1.0.4), and this category is equipped with the Grothendieck topology which is the restriction of the one (Definition 1.0.5) on Prof to Prof_κ. (♠ TODO: restriction of a site to a subcategory).

It should be noted that Prof_κ is an essentially small site, unlike Prof.

Definition 1.0.7 ([CS22, Definition 2.4.])**.** Let κ be a cardinal. A *κ-condensed set* is a sheaf of sets on Prof_κ (Definition 1.0.6).

The category of κ-condensed sets is denoted by CondSet_κ.

Explicitly, a κ-condensed set X is a functor

$$X : \text{Prof}_\kappa^{op} \to \text{Set}$$

such that

1. If $(S_i)_{i\in I}$ is a finite collection of objects of Prof_κ^{op}, then

$$X\left(\sqcup_{i\in I} S_i\right) \xrightarrow{\sim} \prod_{i\in I} X\left(S_i\right);$$

2. If $f : T \to S$ is a surjective map in Prof_κ, then

$$X(S) \overset{f^*}{\hookrightarrow} X(T)$$

is injective with image equal to the set of those $x \in X(T)$ for which $p_1^* x = p_2^* x$, where $p_1, p_2 : T \times_S T \to T$ are the two projections.

(♠ TODO: talk about how the category of extremely disconnected sets can be used to alternatively define condensed sets.)

Definition 1.0.8. Let $\mathcal{A}$ be a category. Let κ be a cardinal

A *κ-condensed object of $\mathcal{A}$* is a sheaf on Prof_κ (Definition 1.0.8) valued in $\mathcal{A}$.

Names for objects of $\mathcal{A}$ induce names for κ-condensed objects of $\mathcal{A}$, e.g. a *κ-condensed group* is a κ-condensed object of the category of groups, etc.

As per Proposition A.0.1, for γ a "kind of algebraic structure defined by finite projective limits", the category of κ-condensed objects of $\gamma - \mathbf{Sets}$ is equivalent to the category of γ-objects of $\mathrm{CondSet}_\kappa$ (Definition 1.0.7).

Definition 1.0.9 ([CS22, After Definition 2.4.])**.** The *category of condensed sets*, denoted CondSet, is the colimit category (♠ TODO: define limit/colimit categories)

$$\mathrm{CondSet} \coloneqq \varinjlim_{\kappa} \mathrm{CondSet}_\kappa$$

(Definition 1.0.6) over all cardinals κ, where the functors $\mathrm{CondSet}_\kappa \to \mathrm{CondSet}_{\kappa'}$ for $\kappa < \kappa'$ are the pullback functors of the inclusion functors $\mathrm{Prof}_\kappa \hookrightarrow \mathrm{Prof}_{\kappa'}$. (♠ TODO: pullback functors)

This category is generated under colimits by arbitrary profinite sets (♠ TODO: explain more on the yoneda embedding)

Definition 1.0.10. Let $\mathcal{A}$ be a category.

A *condensed object of $\mathcal{A}$* is an object (♠ TODO: the intent here is for us to not say κ.)

The category of condensed objects of $\mathcal{A}$ may be denoted as $\mathrm{Cond}(\mathcal{A})$. For example, $\mathrm{Cond}(\mathrm{Ab})$ denotes the category of condensed abelian groups .

Definition 1.0.11. Given a topological space T, define its *associated condensed set* $\underline{T}$ as the functor (♠ TODO: technically, do I define it on Prof kappa or Prof?)

$$\mathrm{Prof}^{\mathrm{op}} \to \mathbf{Sets}, \quad S \mapsto C(S,T),$$

where $C(S,T)$ is the set of continuous maps from S to T. (♠ TODO: talk about how the yoneda embedding of profinite sets into the category of condensed sets is given in the same way.)

Note that if T has the structure of a topological group, topological ring, etc., then $\underline{T}$ is condensed group (Definition 1.0.10), condensed ring, etc.

2. Solid condensed abelian groups

Definition 2.0.1 ([Sch19, Before Definition 2.4.]). Let $S = \varprojlim_i S_i$ be a profinite set (Definition 1.0.1) with S_i finite discrete sets.

Define

$$\mathbb{Z}[S]^{\blacksquare} := \varprojlim \mathbb{Z}[S_i]$$

where $\mathbb{Z}[S_i]$ is the free ring generated by S_i and the morphisms among these is the natural morphism between the free rings induced by the morphisms of finite sets. It is appropriate to call $\mathbb{Z}[S]^{\blacksquare}$ the *solidification of the free ring $\mathbb{Z}[S]$ as a conddensed abelian group* (Definition 1.0.10) — indeed, see (♠ TODO:)

Definition 2.0.2 ([Sch19, Definition 2.4.]). A condensed abelian group (Definition 1.0.10) is solid if for any profinite set (Definition 1.0.1) S with a map $f : S \to M$, there is a unique map $\widetilde{f} : \mathbb{Z}[S]^{\blacksquare} \to M$ (Definition 2.0.1) such that the composition (♠ TODO: talk about yoneda embedding so that S makes sense as a condensed set)

$$S \to \mathbb{Z}[S]^{\blacksquare} \xrightarrow{\widetilde{f}} M$$

is f.

The full subcategory of Cond(Ab) (Definition 1.0.10) consisting of solid objects may be denoted by Solid or Sol.

Theorem 2.0.3 ([?, Theorem 5.8.(i), Theorem 6.2.(i)], see also [Sch19, Theorem 2.5.]).
1. The category Solid of solid abelian groups is closed under small limits, small colimits, and extensions as a subcategory of the category Cond(Ab) (Definition 1.0.10).
2. The inclusion functor Solid $\hookrightarrow$ Cond(Ab) admits a left adjoint, denoted by $M \mapsto M^{\bullet}$ and called *solidification*. As an endofunctor of Cond(Ab), the solidification functor commutes with all colimits and takes $\mathbb{Z}[S]$ to $\mathbb{Z}[S]^{\bullet}$ (Definition 2.0.1).
3. The category of solid abelian groups admits compact projective generators, which are exactly the condensed abelian groups of the form $\prod_I \mathbb{Z}$ for sets I.
4. There is a unique symmetric monoidal tensor product, denoted by $\otimes^{\blacksquare}$, on Solid, making the solidification functor Cond(Ab) $\to$ Solid : $M \mapsto M^{\blacksquare}$ symmetric monoidal. (♠ TODO: what is the symmetric monoidal functor on CondAb then?)

3. Solid modules

Definition 3.0.1. Let R be a topological ring.

A *solid R-module* is a solid condensed abelian group (Definition 2.0.2) that is also a module of $\underline{R}$ (Definition 1.0.11) (♠ TODO: should have to think — given a condensed abelian group, what site is it a sheaf on, and what is underline a sheaf on?)

(♠ TODO: what are morhpisms)

The category of solid R-modules may be denoted by Solid_R or Sol_R.

Given two topological rings R and S, we may accordingly define the notion of *solid R-S-bimodule*. The category of such objects may be denoted by ${}_R\,\mathrm{Solid}_S$ or ${}_R\,\mathrm{Sol}_S$.

(♠ TODO: solid R-sheaf on a scheme)

(♠ TODO: proetale site) (♠ TODO: frobenius operator on)

Appendix A. Miscellaneous

Proposition A.0.1 ([GV72, Exposé II, Proposition 6.3.1])**.** Let $\mathcal{C}$ be a site, and let γ be a "kind of algebraic structure defined by finite projective limits" (♠ TODO: Probably the correct terminology for this is "essentially algebraic categories" or "Lawvere theories") (e.g. γ can be the notion of group objects or abelian group objects, ring objects, or module objects). The following categories are equivalent:

1. The category of sheaves on $\mathcal{C}$ with values in the category of γ-sets,
2. The category of γ-objects of the category of sheaves of sets on $\mathcal{C}$, and
3. The category of γ-objects of the category of presheaves of sets on $\mathcal{C}$ whose underlying presheaf is a sheaf of sets.

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